A1. $0 = U_d + p\omega_m L_q I_q$

$\Rightarrow U_d = -p\omega_m L_q I_q$

A2.

$$\left.\begin{aligned} L_q \frac{dI_q}{dt} &= U_q - RI_q - p\omega_m \psi_{PM} \\ \hat{U}_q &= L_q \frac{dI_q}{dt} + RI_q \\ U_q &= \hat{U}_q + U_{q,komp} \end{aligned}\right\} \Rightarrow U_{q,komp} = p\omega_m \psi_{PM}$$

A3.1

$$\hat{U}_q = L_q \frac{dI_q}{dt} + RI_q$$

$$\hat{U}_q(s) = (L_q s + R) I_q(s) \quad \downarrow \text{Laplace-Transformation}$$

$$H_{el}(s) = \frac{I_q(s)}{\hat{U}_q(s)} = \frac{1}{sL_q + R} = \frac{1/R}{1 + s(L_q/R)}$$

$\Downarrow$ PT1-System

$V_s = 1/R$

$T_{el} = L_q/R$

A3.2

$$H_{LV}(s) = \frac{\hat{U}_q(s)}{\hat{U}_{q,ideal}(s)} = \frac{1}{1 + sT_{LV}}$$

gain = 1

$T = T_{LV}$ time constant

A3.3

$$G_I(s) = H_{LV}(s) \cdot H_{el}(s) = \frac{1}{1+sT_{LV}} \cdot \frac{1/R}{1+s(L_q/R)} = \frac{1/R}{(1+sT_{LV})(1+sT_{el})} = \frac{1}{(L_q s + R)(T_{LV} s + 1)}$$

$\Rightarrow$ PT2-Glied

A3.4

Typ: PI BO

$T_1 = T_{el} = \frac{L_q}{R}$ $T_6 = T_{LV}$ $V_S = 1/R$

$$V_{IR} = \frac{T_1}{2T_6 V_S} = \frac{T_{el}}{2 \cdot T_{LV} \cdot (1/R)} = \frac{R \cdot T_{el}}{2T_{LV}}$$

$T_{In} = T_1 = T_{el}$

$$R_I(s) = V_{IR} \frac{1 + sT_{In}}{sT_{In}} = \frac{R \cdot T_{el}}{2T_{LV}} \cdot \frac{1 + sT_{el}}{sT_{el}}$$

A3.5



A3.6

$$K_I(s) = \frac{R_I(s) \cdot G_I(s)}{1 + R_I(s) \cdot G_I(s)} = \frac{T_{el} \cdot (1 + sT_{el})}{2(T_{el}s + 1)(T_{LV}s + 1)T_{LV} \cdot sT_{el} + T_{el} \cdot (1 + sT_{el})}$$

$$= \frac{1}{2s^2T_{LV}^2 + 2sT_{LV} + 1} \Rightarrow \text{PT2-Glied}$$

A4.1

$$J\frac{d\omega_m}{dt} = M_m - M_{Lm}^{\,0} \qquad M_m = \frac{3}{2}p\Psi_{PM}I_q$$

$$H_w(s) = \frac{\omega_m(s)}{I_q(s)} = \frac{\frac{3}{2}p\Psi_{PM}}{Js} \Rightarrow \text{I-Glied}$$

A4.2

$$G_w(s) = H_w(s) \cdot K_I(s) \qquad K_I(s) \overset{\text{Approach}}{=} \frac{1}{1 + s \cdot 2T_{LV}}$$

$$= \frac{\frac{3}{2}p\Psi_{PM}}{J \cdot s} \cdot \frac{1}{1 + s \cdot 2T_{LV}}$$

$\Rightarrow$ IT1-Glied

A4.3

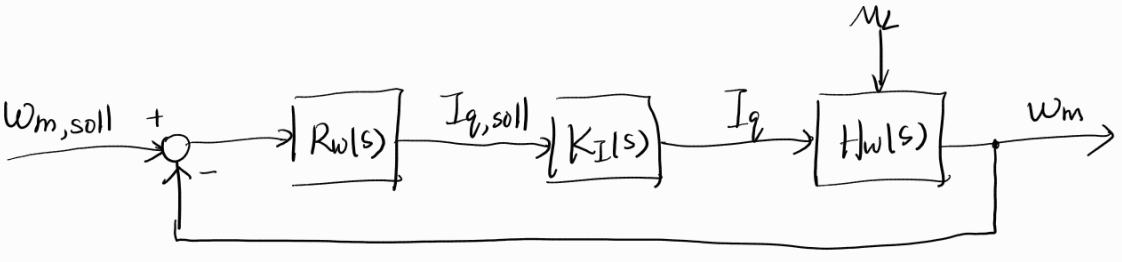
$$\text{Typ: PI} \quad , \quad SO \quad := \quad T_6 = 2T_{LV} \quad , \quad \frac{V_S}{T_1} = \frac{\frac{3}{2}p\Psi_{PM}}{J}$$

$$V_{WR} = \frac{T_1}{2T_6V_S} = \frac{1}{2(2T_{LV})}\left(\frac{J}{\frac{3}{2}p\Psi_{PM}}\right) = \frac{J}{6T_{LV}\,p\Psi_{PM}}$$

$$T_{WN} = 4T_6 = 4 \cdot 2T_{LV} = 8T_{LV}$$

$$R_w(s) = V_{WR} \cdot \frac{1 + sT_{WN}}{sT_{WN}} = \frac{J}{6T_{LV} \cdot p\Psi_{PM}} \cdot \frac{1 + 8T_{LV} \cdot s}{8T_{LV} \cdot s}$$

A4.4



A4.5

$$K_w(s) = \frac{G_w(s) \cdot R_w(s)}{1 + G_w(s) \cdot R_w(s)} = \frac{1 + 8sT_{LV}}{32s^2T_{LV}^2(1 + 2sT_{LV}) + (1 + 8sT_{LV})}$$

A5.1

$$\omega_m = \frac{d\varphi_m}{dt} \Rightarrow H_\varphi(s) = \frac{\varphi_m(s)}{\omega_m(s)} = \frac{1}{s} \Rightarrow \text{I Glied}$$

A5.2

$$G_\varphi(s) = K_w(s) \cdot H_\varphi(s) \qquad K_w(s) \overset{\text{PT1-element}}{\approx} \frac{1}{1 + s \cdot 8T_{LV}}$$

$$= \frac{1}{1 + s \cdot 8T_{LV}} \cdot \frac{1}{s} \Rightarrow \text{IT1-Glied}$$

Variante 1: Typ: P, BO $\quad V_R = \frac{T_1}{2T_6 V_S}$

mit $T_1 = 1 \quad T_6 = 8T_{LV}, \quad V_S = 1$

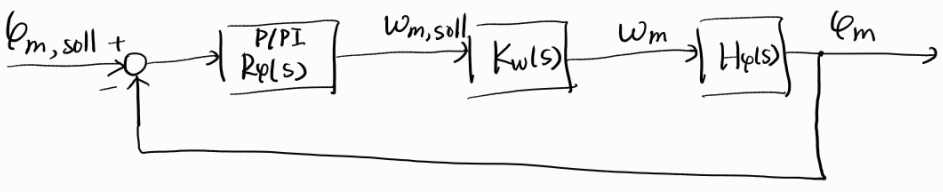
$$V_{\varphi R}^P = \frac{T_1}{2T_6 V_S} = \frac{1}{2T_6}\left(\frac{T_1}{V_S}\right) = \frac{1}{16T_{LV}} \qquad R_\varphi^P(s) = \frac{1}{16T_{LV}}$$

Variante 2: Typ: PI, SO $\quad T_n = 4T_6 \quad V_R = \frac{T_1}{2T_6 V_S}$

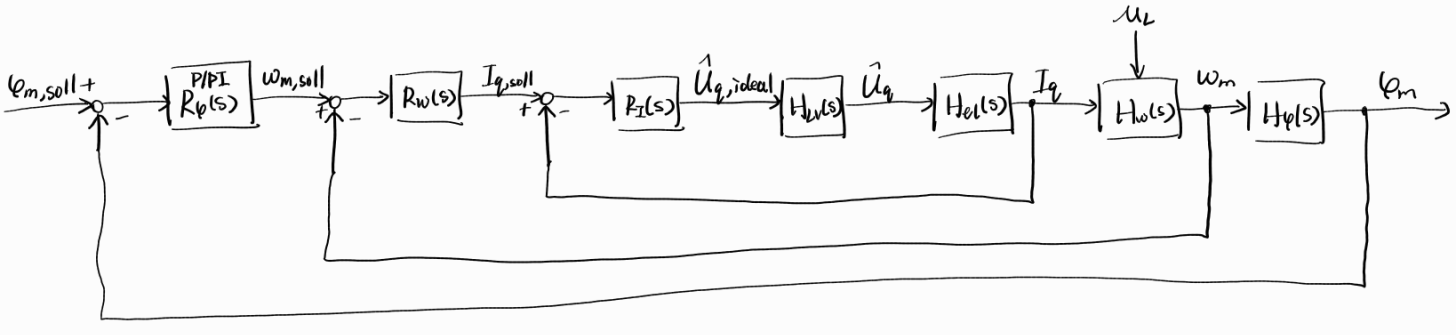
$$V_{\varphi R}^{PI} = \frac{T_1}{2T_6 V_S} = \frac{1}{16T_{LV}}, \quad T_{\varphi n}^{PI} = 4T_6 = 32T_{LV}$$

$$R_\varphi^{PI}(s) = \frac{1}{16T_{LV}} \cdot \frac{1 + s \cdot 32T_{LV}}{s \cdot 32T_{LV}}$$

A5.3



A6



A7.2

Observation: P-controller has a constant steady-state error. With the PI-controller, this steady-state error is eliminated.

Justification: This is due to the integral (I) component. The I part integrates the error over time, continuously increasing the control output until the error is completely compensated.

A7.3 noise

The measurement noise leads to the high-frequency oscillations. This would cause vibrations on real hardware.

How to improve = 1. Apply A low-pass filter
2. Adjusting the PID parameters.

A7.4 $r_abs = 5e^{-5}$ noise ↑

Observation = The system becomes completely unstable and cannot follow the setpoint.

Critical value of r_abs is $r_abs \approx 3.77e^{-5}$

A7.5

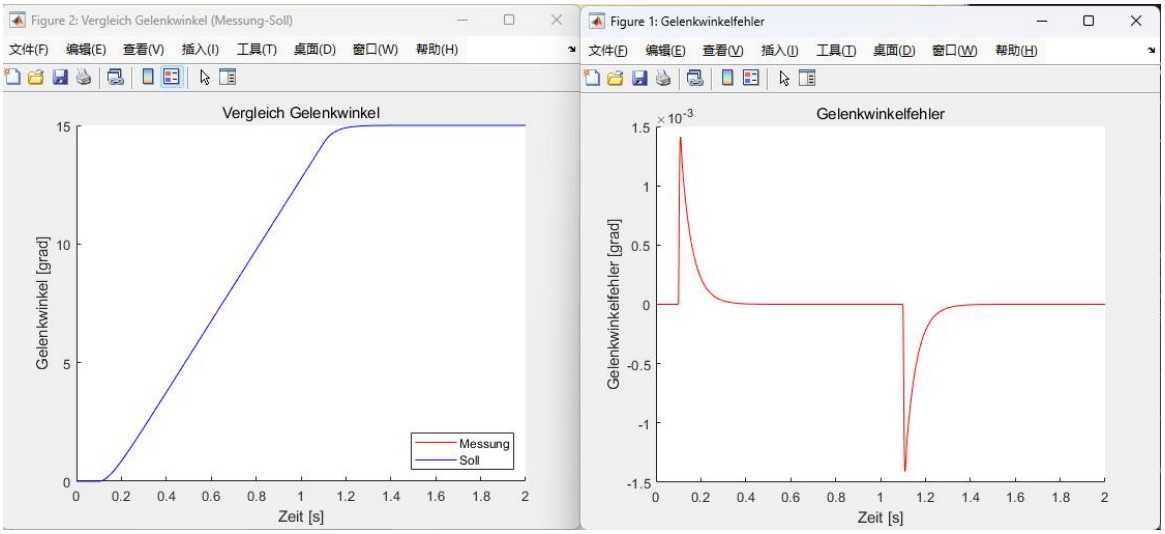
A step input is physically impossible, because it requires extreme acceleration. The controller tries to chieve this by commanding maximum output. which causes actuator saturation. This saturation breaks the linear model, leading to large overshoot and instability.

Variante = 2; % PI-Positionsregler

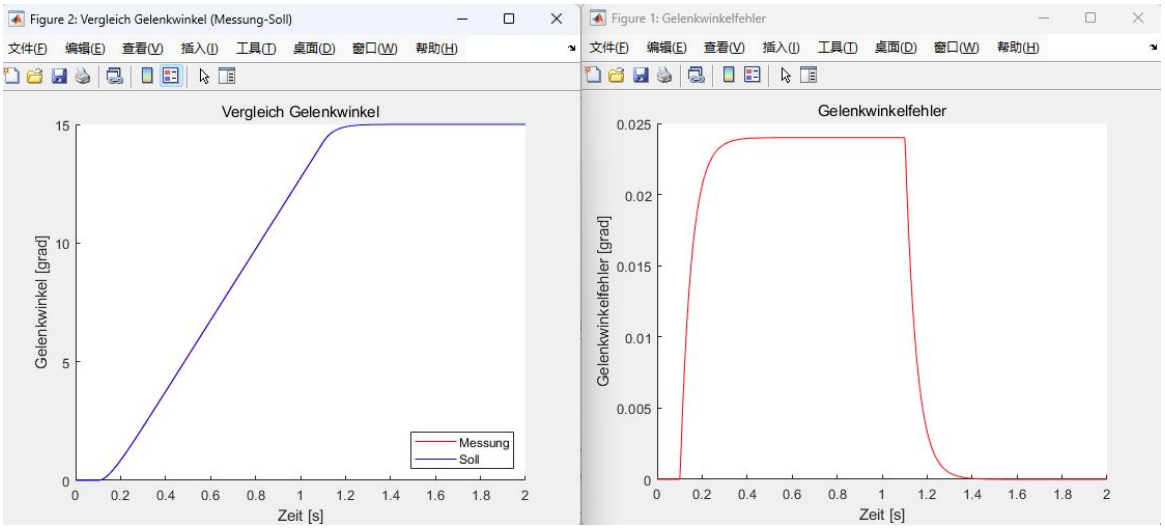
Last = true; % Mit Last

Messrauschen = false;

Signaltyp = 2;



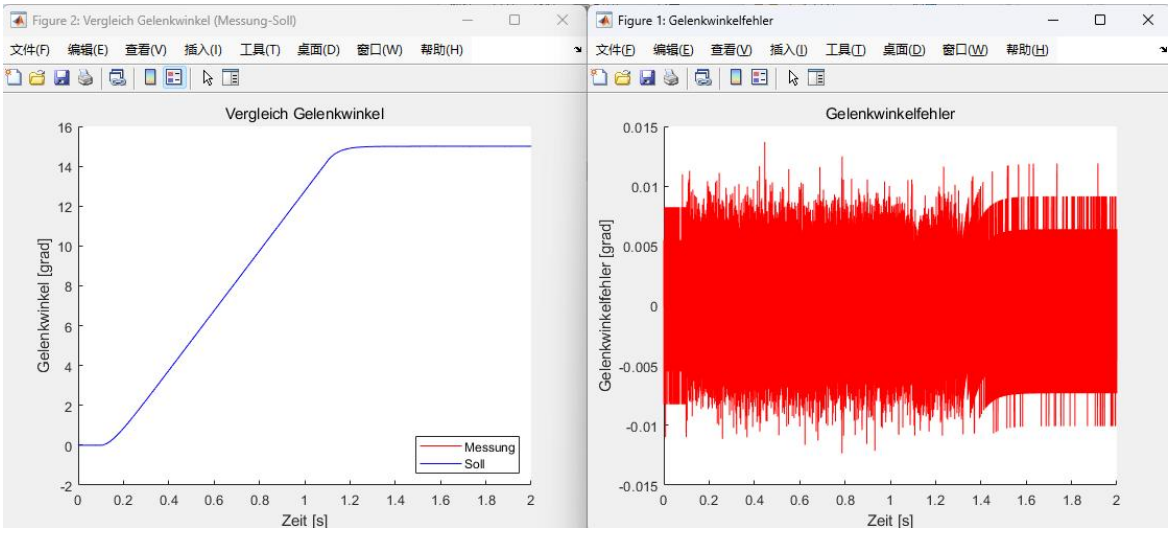
Variante = 1;



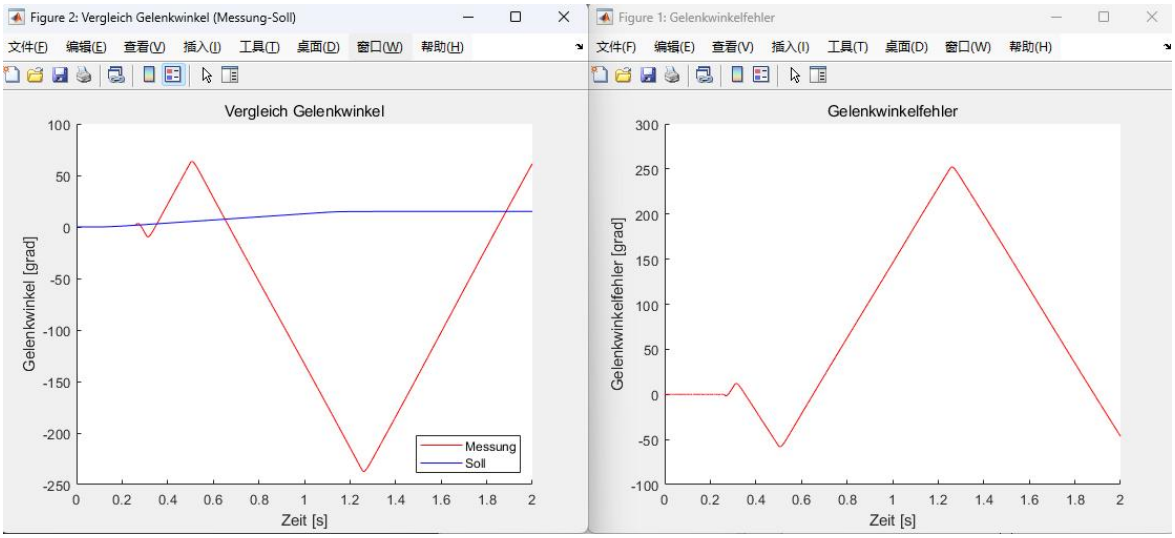
Variante = 2; % PI-Positionsregler

Last = true;

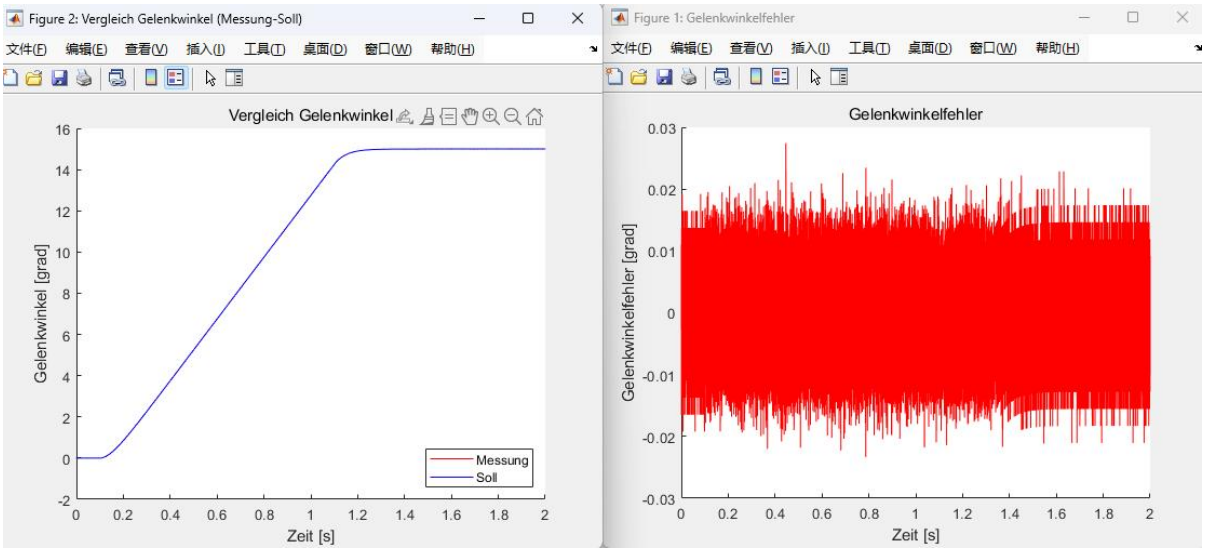
Messrauschen = true;



r_abs=5e-5



Critical value r_abs=5e-5



Variante = 2; % PI-Positionsregler

Last = true; % Mit Last

Messrauschen = false;

Signaltyp = 1; %sprung

